COMP 3004 FINAL PROJECT

Note: Users refer to RaDoTech Customers/clients who use the device to do a health scan

Use Case 1: Initial Setup/Onboarding of RaDoTech Device

Primary Actors: Users, RaDoTech Device suppliers

Stakeholders and Interest: Users -> follow instructions clearly, download the app, set up the device as well as complete their initial user profile.

RaDoTech Suppliers -> ensure device has reached user successfully.

Precondition: RaDoTech Device, cable and spray bottle have been received.

Success Guarantee: App downloaded for any time use, and User profile has been created successfully.

Main Success Scenario:

1. Download the RaDoTech app from apple store or play store depending on system OS.
2. User first time using app, click on "Create Profile"
3. Enter all user information and verify if it's correct
4. Click "Save and Continue"
5. click triple line icon and go to profiles and verify profile has been created successfully.

Extensions:

5a) profile not shown up under saved profiles

5a1) Close the app and retry creating profile again, make sure not to leave anything blank.

Use Case 2: Preparing/Setting Up for first scan

Primary Actors: Users

Stakeholders and Interest: Users -> follow instructions clearly set up organ for first scan and pair device.

Precondition: remove metal jewelry/watch from body.

Success Guarantee: Device paired with app and scan site sprayed with water ready for first scan.

Main Success Scenario:

1. Click "Measure now" on the app
2. Replicate hand posture displayed in the tips section, hold device overhand with palm resting on the metal bar of the device, do not hold it like a pencil. Click next.
3. Use the spray bottle that came alongside the device and spray the area user wishes to scan, spray one time to moisturize area. Click close.
4. Press and hold power button of RaDoTech device and it will flash blue, pair the device with the RaDoTech app. Click on "Start Metering"
5. Select User profile and you will be redirected to screen showing the lines to scan along on the organ

Extensions:

3a) spray bottle empty or ran out of liquid

3a1) spray bottle is just tap water, fill the bottle again and continue to use.

4a) No blue Flashing displayed on RaDoTech Device.

4a1) RaDoTech Device Battery ran out of charge, please charge the device and try again.

Use Case 3: Start Metering/Scanning

Primary Actors: Users

Stakeholders and Interest: Users -> follow onscreen instructions and follow device node through the lines displayed on the screen tracing the scanning path on the organ.

Precondition: Jewelry removed from body, User profile selected, Scan site of organ has been sprayed with water RaDoTech Device paired with app.

Success Guarantee: Successfully scanned all vital points of each organ for an accurate scan result.

Main Success Scenario:

1. Hold RaDoTech device overhand and place the node of the device on the meridian (vital point displayed on the screen of the app) and hold it there until you hear a beep sound.
2. Repeat step 1 for each meridian on the organ as displayed by the app.
3. Repeat steps 1 and 2 for each organ as indicated by the app and follow through until you have completed all the organs and are directed to the "Notes" Screen.
4. On the notes section User can enter how they were feeling during the scan and any other information they would like to record. Click "Save" after completing notes.

Extensions:

1a) Cannot hear a beep sound, scanned the first meridian but app does not move onto the next.

1a1) Check app for low battery indication, device might be on low charge and needs to be charged again before use.

1a2) RaDoTech Device did not pair properly, re-pair app with device.

3a) Some organs like feet are hard to reach using the device, unable to scan.

3a1) RaDoTech Device can be extended from the top of the metal bar to provide more length to enable scanning of hard to reach meridians.

Use Case 4: Viewing Results of Scan

Primary Actors: Users

Stakeholders and Interest: Users -> after completing the scan users can see their results in 3 ways on the "Metering results" page.

Precondition: completed scan of all meridians on all the organs indicated by the app.

Success Guarantee: Users will be able to view all the organs' information based off the scanned meridians And indicate to the user which organ is performing below/above average.

Main Success Scenario:

1. After completing scan go back to home screen of the app.
2. Click on the triple line icon, go to "History" and select the scan performed on today's date.
3. The first screen is the "Body" screen which shows a body shot of the scanned organs and is colour coded in green, blue and red.
4. The colours indicate the status of each organ when green means normal, blue means below normal and red means above normal.
5. The second screen is the "Chart" screen shows what the results mean at an organ level displaying values of the scan to show performance of each organ.
6. The third screen is the "Circle" screen that puts the data altogether and compares all the organs performances by comparing the right and left side of the meridians and comparing it with the average values/normal readings you have.
7. Clicking on the "Indicators" tab displays the screen that shows snapshot of how your systems are overall performing and gives users info varying from energy levels to psycho-emotional state.
8. Clicking "Recommendations" tab shows the screen that will provide a snapshot with more insight regarding the information from the indicators screen and provide the user with actionable tips on how to bring abnormal levels of any organ back to normal state. This screen also displays the user's type of lifestyle if they are too hyperactive or lazy etc.

Extensions:

4a) User wants to bring back organs to normal level/make them all green coloured.

4a1) Head to the recommendations tab and check for advice on how to bring back above/below performing organs to their normal state.

7a) User does not understand what the tabs mean or represent.

7a1) clicking on the tab will expand that section providing more detailed information on how this result came to be and what state the organ is in.

Use Case 5: Contacting RaDoTech Team

Primary Actors: Users, RaDoTech Support Team

Stakeholders and Interest: Users -> Would like to contact RaDoTech Support team for more information.

RaDoTech Support Team -> Respond to the Users questions and provide accurate, actionable feedback.

Precondition: User has already taken a previous scan and was able to view their results.

Success Guarantee: Users request gets answered successfully and correctly by the RaDoTech Team.

Main Success Scenario:

1. User wants to contact RaDoTech Team but app does not have a means to reach out.
2. User should email prime@radotek.com for help or if they would like to thank the team.

Extensions:

2a) User wishes to request more supplements.

2a1) RaDoTech team receives the email and responds to User indicating which supplements are available and by when it will be delivered to the user .

2b) User requests clarity on results

2b1) RaDoTech responds to user question and provide a more in-depth explanation and may request additional scans or information from the user to provide better feedback.

Main Use Case: User takes their first scan

Primary Actors: Users, RaDoTech Device suppliers, RaDoTech Support team

Stakeholders and Interest: Users -> follow instructions clearly, download the app, set up the device as well as complete their initial user profile.

RaDoTech Suppliers -> ensure device has reached user successfully.

RaDoTech Support Team -> Respond to the Users questions and provide accurate, actionable feedback.

Precondition: RaDoTech Device, cable and spray bottle have been received; jewelry removed from body.

Success Guarantee: User Successfully takes their first scan and can see the results on the app.

Main Success Scenario:

1. Download the RaDoTech app from apple store or play store depending on system OS. (see Use 1)
2. User first time using app, click on “Create Profile” (see Use case 1)
3. Enter all user information and verify if it’s correct (see Use case 1)
4. Click “Save and Continue” (see Use case 1)

5. click triple line icon and go to profiles and verify profile has been created successfully. (see Use case 1)
6. Click “Measure now” on the app (see Use case 2)
7. Replicate hand posture displayed in the tips section, hold device overhand with palm resting on the metal bar of the device, do not hold it like a pencil. Click next.(See use case 2)
8. Use the spray bottle that came alongside the device and spray the area user wishes to scan, spray one time to moisturize area. Click close. (See Use case 2)
9. Press and hold power button of RaDoTech device and it will flash blue, pair the device with the RaDoTech app. Click on “Start Metering” (see Use case 2)
10. Select User profile and you will be redirected to screen showing the lines to scan along on the organ (see Use case 2)
11. Hold RaDoTech device overhand and place the node of the device on the meridian (vital point displayed on the screen of the app) and hold it there until you hear a beep sound.
12. Repeat step 1 for each meridian on the organ as displayed by the app.
13. Repeat steps 1 and 2 for each organ as indicated by the app and follow through until you have completed all the organs and are directed to the “Notes” Screen.
14. On the notes section User can enter how they were feeling during the scan and any other information they would like to record. Click “Save” after completing notes.
15. After completing scan go back to home screen of the app.
16. Click on the triple line icon, go to “History” and select the scan performed on today’s date.
17. The first screen is the “Body” screen which shows a body shot of the scanned organs and is colour coded in green, blue and red.
18. The colours indicate the status of each organ when green means normal, blue means below normal and red means above normal.
19. The second screen is the “Chart” screen shows what the results mean at an organ level displaying values of the scan to show performance of each organ.
20. The third screen is the “Circle” screen that puts the data altogether and compares all the organs performances by comparing the right and left side of the meridians and comparing it with the average values/normal readings you have.
21. Clicking on the “Indicators” tab displays the screen that shows snapshot of how your systems are overall performing and gives users info varying from energy levels to psycho-emotional state.
22. Clicking “Recommendations” tab shows the screen that will provide a snapshot with more insight regarding the information from the indicators screen and provide the user with actionable tips on how to bring abnormal levels of any organ back to normal state. This screen also displays the user’s type of lifestyle if they are too hyperactive or lazy etc.
23. User wants to contact RaDoTech Team but app does not have a means to reach out.
24. User should email prime@radotek.com for help or if they would like to thank the team.

Extensions:

5a) profile not shown up under saved profiles

5a1) Close the app and retry creating profile again, make sure not to leave anything blank.

(See Use Case 1, Extension 5a)

8a) spray bottle empty or ran out of liquid

8a1) spray bottle is just tap water, fill the bottle again and continue to use.

(Use case 2 Extension 3a)

9a) No blue Flashing displayed on RaDoTech Device.

9a1) RaDoTech Device Battery ran out of charge, please charge the device and try again. **(Use case 2 Extension 4a)**

11a) Cannot hear a beep sound, scanned the first meridian but app does not move onto the next.

11a1) Check app for low battery indication, device might be on low charge and needs to be charged again before use **(Use Case 3 Extension 1a1)**

11a2) RaDoTech Device did not pair properly, re-pair app with device**.(Use Case 3 Extension 1a2)**

13a) Some organs like feet are hard to reach using the device, unable to scan.

13a1) RaDoTech Device can be extended from the top of the metal bar to provide more length to enable scanning of hard to reach meridians. **(Use Case 3 Extension 3a)**

18a) User wants to bring back organs to normal level/make them all green coloured.

18a1) Head to the recommendations tab and check for advice on how to bring back above/below performing organs to their normal state. **(Use case 4 Extension 4a)**

21a) User does not understand what the tabs mean or represent.

21a1) clicking on the tab will expand that section providing more detailed information on how this result came to be and what state the organ is in. **(Use Case 4 Extension 7a)**

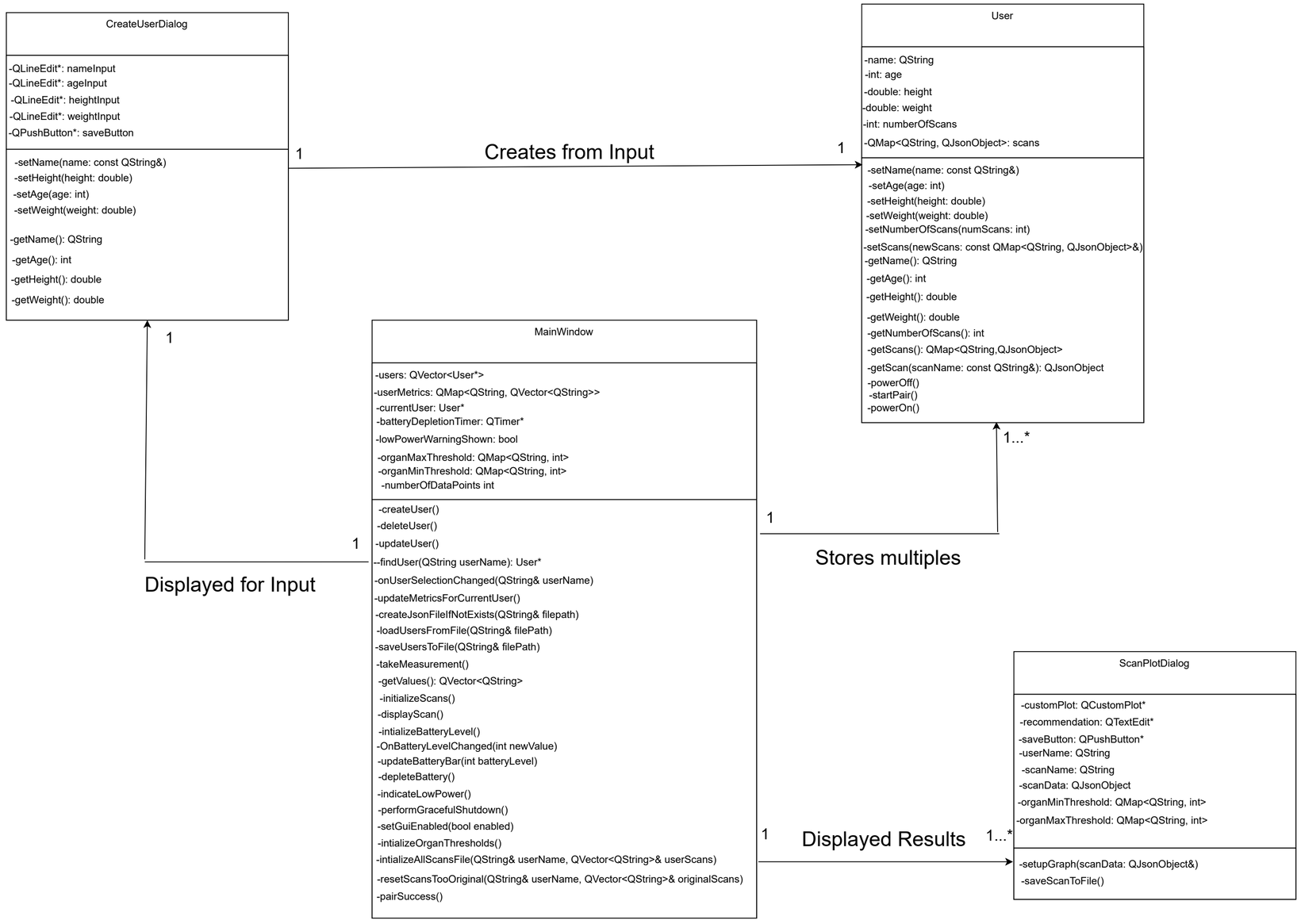
23a) User wishes to request more supplements.

23a1) RaDoTech team receives the email and responds to User indicating which supplements are available and by when it will be delivered to the user. **(Use Case 5 Extension 2a).**

23b) User requests clarity on results

23b1) RaDoTech responds to user question and provide a more in-depth explanation and may request additional scans or information from the user to provide better feedback. **(Use Case 5 Extension 2b)**

UML CLASS DIAGRAM



UML Diagram Design Discussion

CreateUserDialog Class - Contains Input fields (ex: nameInput), User fields these fields and the class sets these as the values for the profile of the User

User Class - represents each user profile, has variables for the user (such as name, height, age) and this is retrieved from the CreateUserDialog Class using getters (ex: getName(), getAge()), the class then sets these values using its setter functions (setName,setAge()) and creates a new User profile with this data, the User class also sets -QMap<QString, QJsonObject>: scans, which is used as a list to store all the scans the user has done in the form of key-value pairs.

MainWindow Class - represents the GUI of the application, User can create a new profile or update his current profile or delete a previous profile. when user creates the profile the application sets up scans for the user using intializeScans(), and once the user goes to the next step of scanning the takeMeasurement() function is called which shows the user which meridians to scan.. After the scan is done updateMetricsForCurrentUser() saves the new rresults on the users profile, and the update profile is saved to a file for the user to download using saveUsersToFile(QString& filePath).

ScanPlotDialog - main class used to show the results to the user in the form of a graph where each reading is recorded as a point. the graph also has two lines representing the min and max thresholds for an organ to show abnormalities if a reading falls below the min or above the max thresholds, this is recording using organMinThreshold and organMaxThreshold.

CreateUserDialog to User class is a 1 : 1 relationship since each dialog is unique to a single user for them to enter information, user cannot have multiple dialogs open for entry at the same time.

MainWindow to CreateUserDialogis a 1 : 1 relationship since 1 user dialog screen is displayed at a time on the GUI, multiple windows are neither stored not displayed.

MainWindow to User is a 1 : N relationship, since the main GUI can showcase multiple user profiles for selection and also store multiple different users

MainWindow to ScanPlotDialog is a 1 : N relationship since the GUI displays multiple screens of resutls (ex: bodyshot, chart, screen which displays measurements at an organ level)

Explaination for Design Decisions

Our code incorporates the concept of persistency by storing user information and scan data in a single users.json file. To meet the requirement of indicating whether the device is in contact with the skin, we modeled this interaction using a button. Each button click represents a single measurement, and after 24 clicks, the system generates a scan. This scan is visualized as a diagram consisting of line graphs.

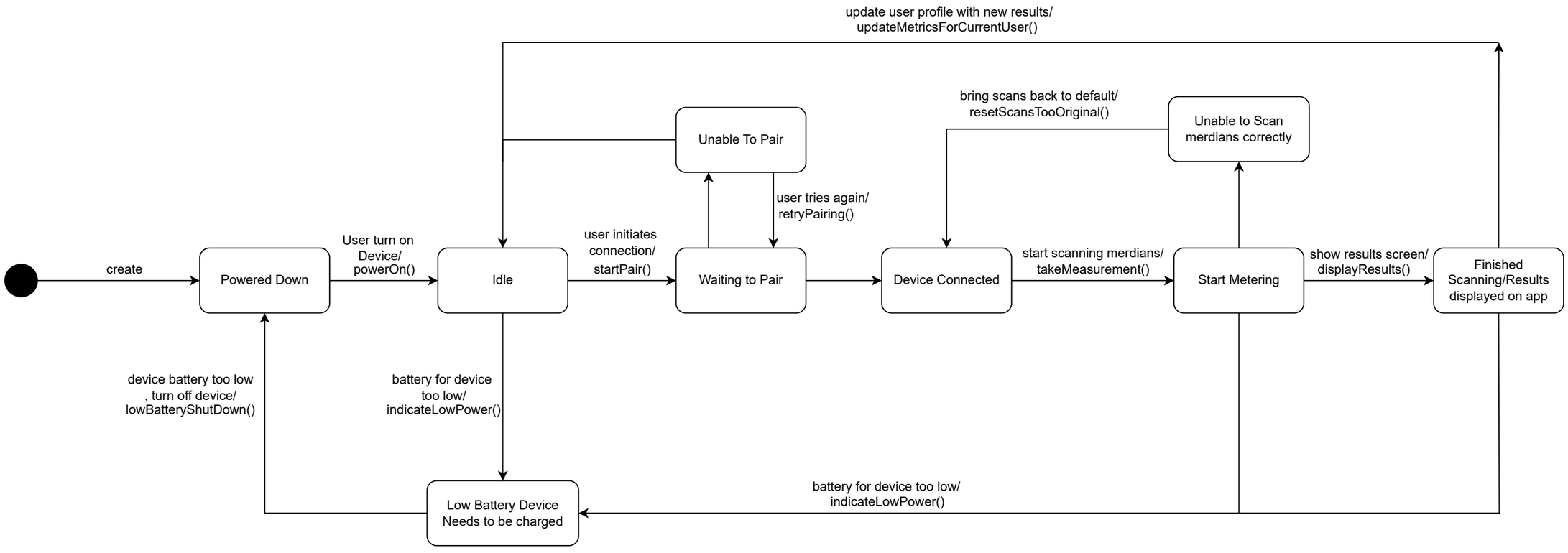
The diagram includes multiple line graphs, where:

- The **green line** represents the maximum acceptable threshold for each organ.
- The **red line** represents the minimum acceptable threshold for optimal health.
- The **blue line** depicts the measured value for the user's scan of a specific organ.

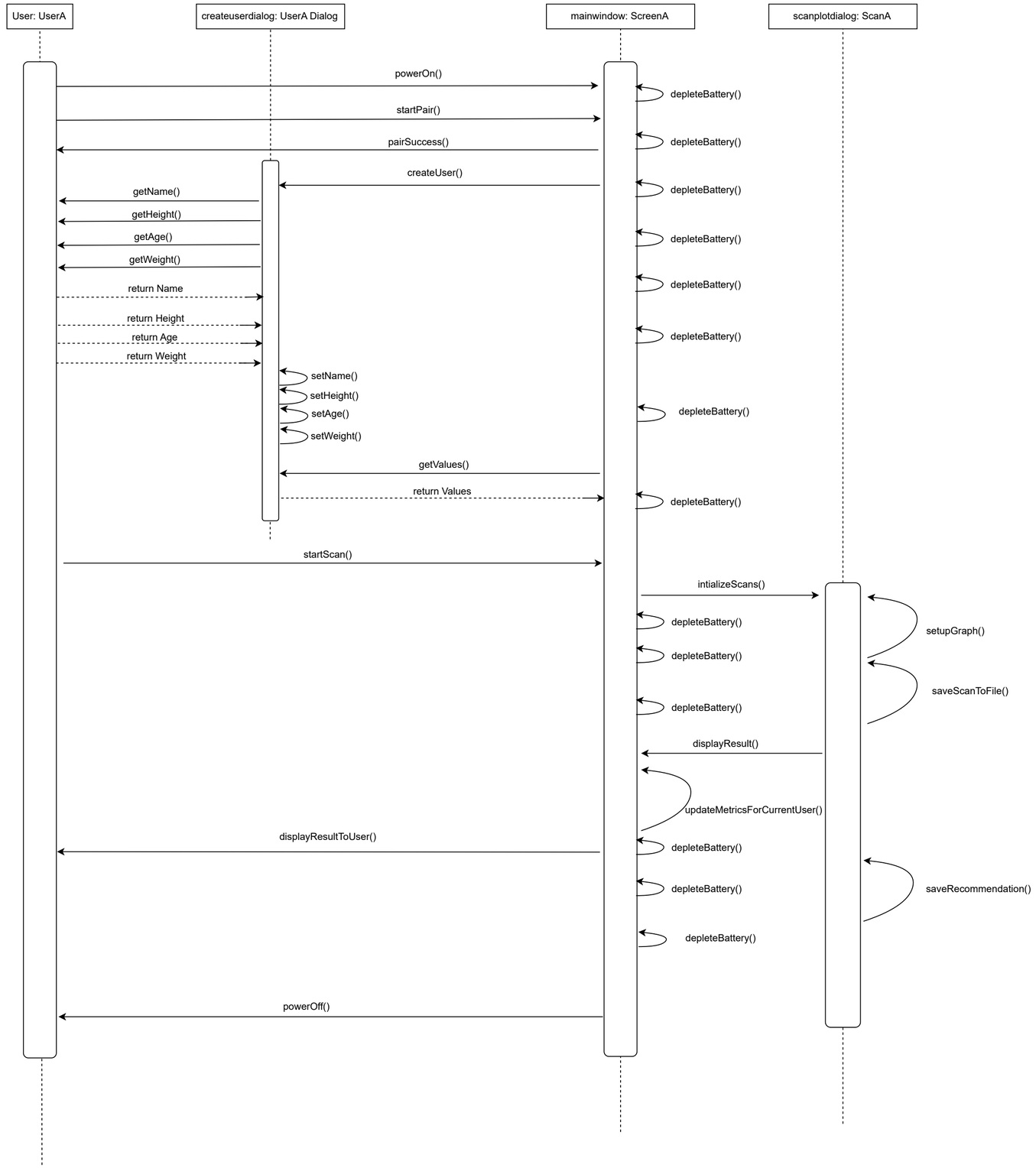
This visualization provides an intuitive and user-friendly simulation of the RadoTech device, making the data easy to interpret. Additionally, each graph includes a text box for personalized recommendations, which are saved alongside the user's scan data in the users.json file.

Other aspects of our code are designed to be intuitive and user-friendly, ensuring a seamless user experience.

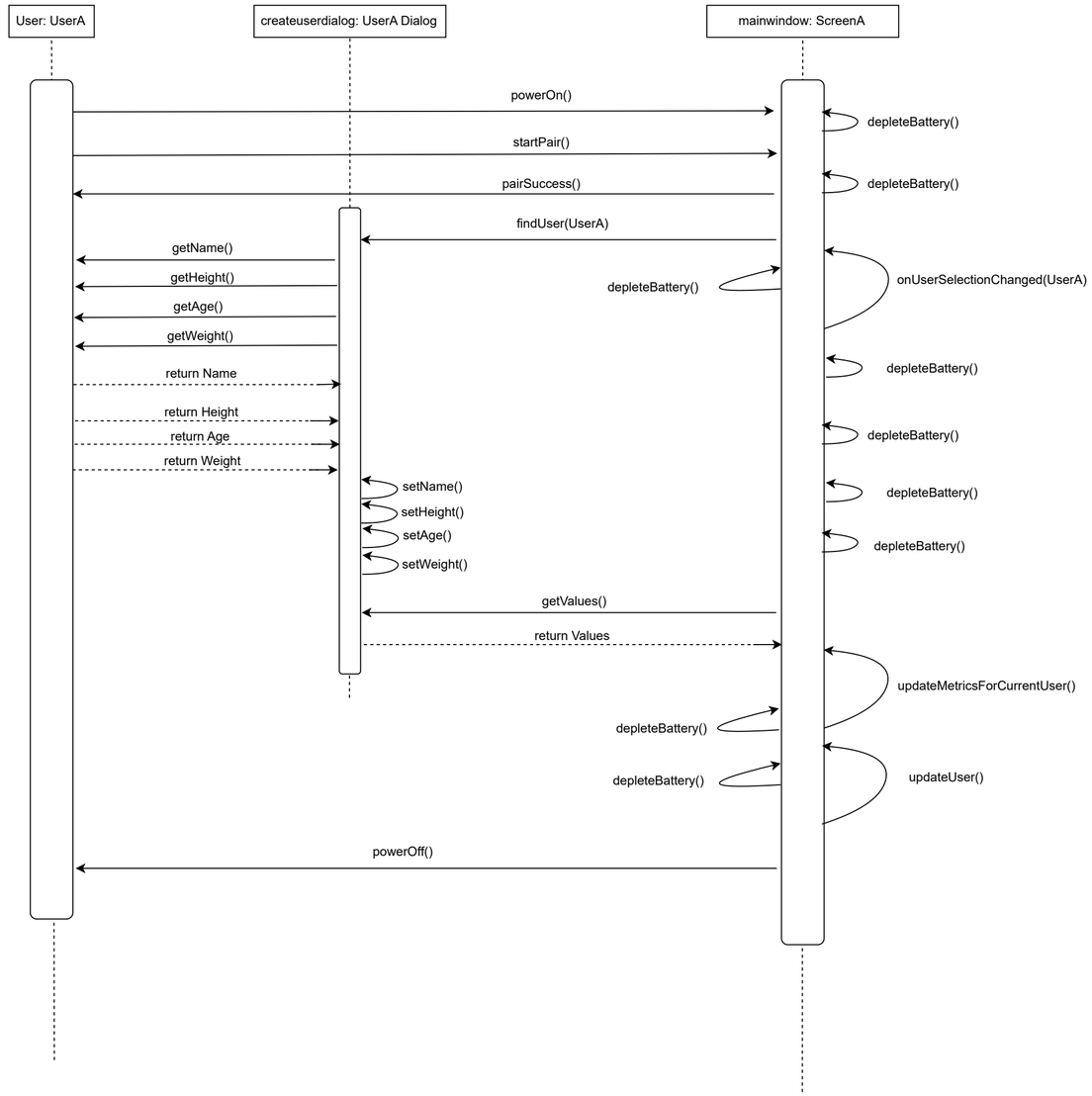
State Machine Diagram for RaDoTech Device



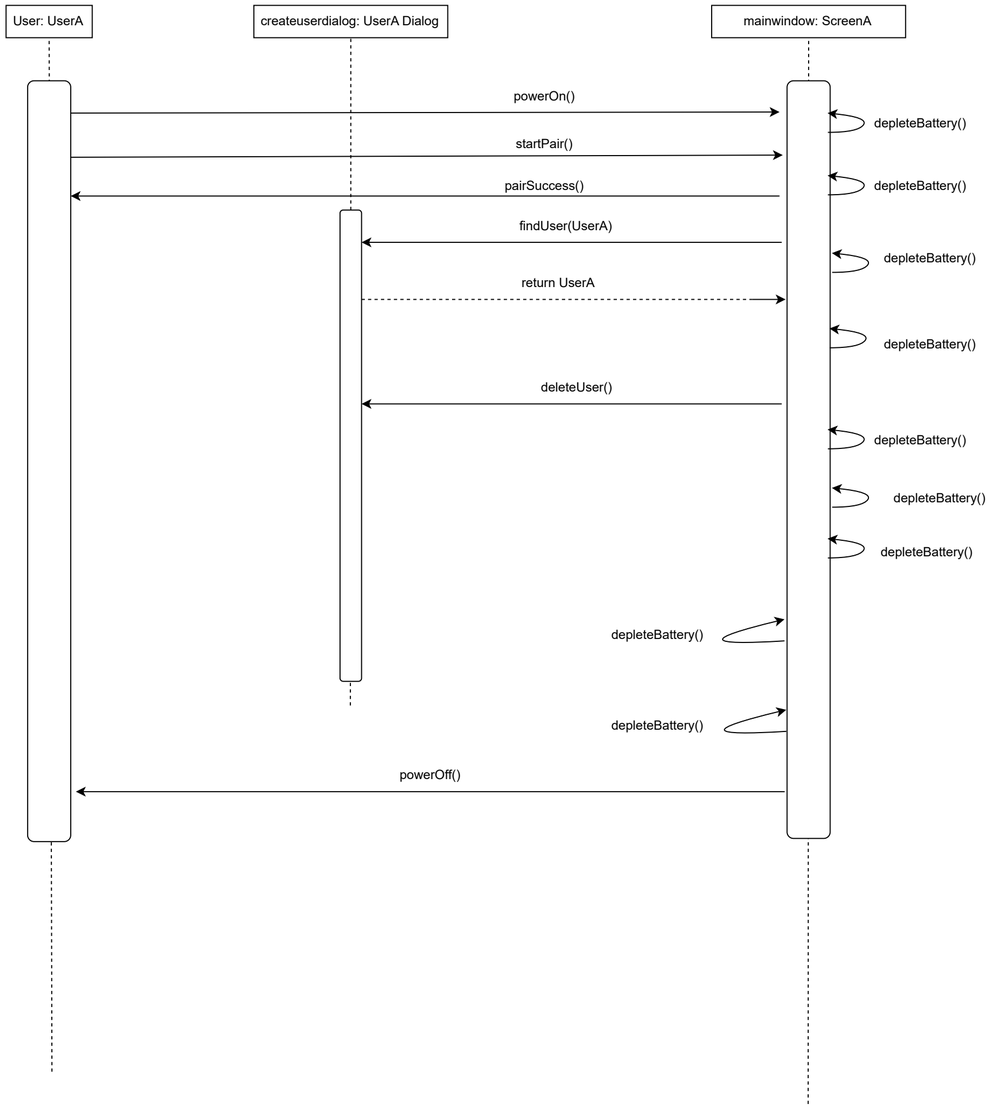
Success Scenario 1
First time user creates profile, pairs the device successfully and takes scans of their organs and the results are displayed to the user.



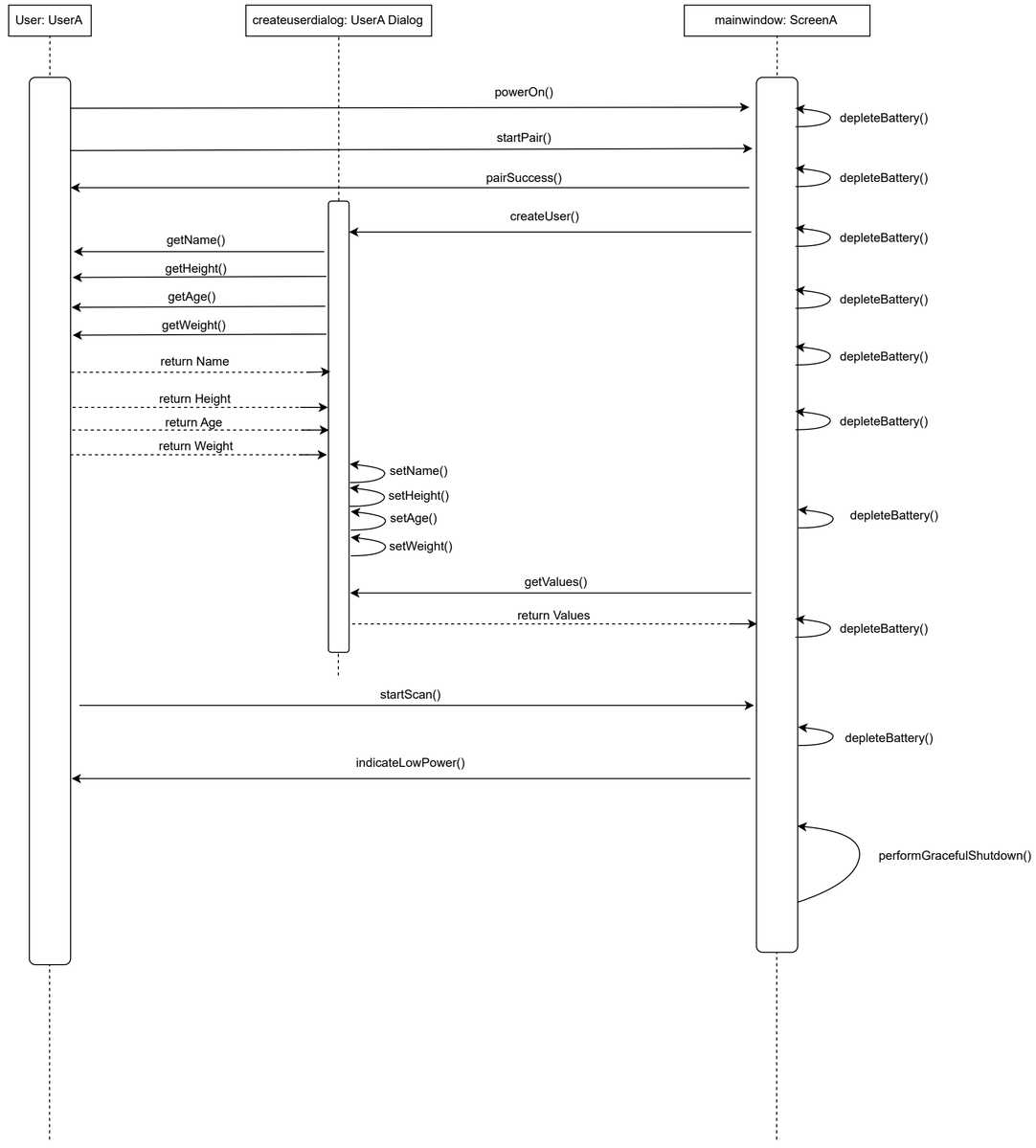
Success Scenario 2
Already Existing User Is updating their user profile. In this case user is changing his personal information such as height, weight, age and name.



Success Scenario 3
Already Existing User wishes to delete their profile and all the data stored in it from the interface.



Safety Scenario 1: Battery too low device shuts down
The User pairs the device successfully and tries to take scan but low battery warning given and device shuts down.



ID	Requirement	Related Use Case	Implemented By	Tested by
1	Users can create a new profile by entering details such as name, age, height, and weight.	Initial Setup/Onboarding	CreateUserDialog, User	Run the application and create a new user profile.
2	User profiles are stored persistently and can be retrieved when the app restarts.	Initial Setup/Onboarding	MainWindow, User	Verify that profiles are saved in the JSON file and loaded on app restart.
3	User profiles are stored persistently and can be retrieved when the app restarts.	Preparing/Setting Up for First Scan	MainWindow	Check the userHistoryComboBox for profile selection.
4	Users can start a new scan for a selected profile.	Preparing/Setting Up for First Scan	MainWindow	Click "Skin Contact" and verify that the scan preparation screen is shown.
5	The app pairs with the RaDoTech device successfully before a scan.	Preparing/Setting Up for First Scan	MainWindow	Observe the pairing process using the takeMeasurement function.
6	The app displays instructions for holding the device correctly and preparing the scan area.	Preparing/Setting Up for First Scan	MainWindow	Verify that the instructions are displayed before the scan begins.
7	Scans are performed by following on-screen instructions to trace meridians of various organs.	Start Metering/Scanning	MainWindow, ScanPlotDialog	Observe the scan flow and the corresponding actions in the app UI.
8	Users can enter notes after completing a scan.	Start Metering/Scanning	ScanPlotDialog	Add notes in the text box and verify they are saved.
9	The system generates and displays graphical results of scans for each organ.	Viewing Results of Scan	ScanPlotDialog, QCustomPlot	Verify graphs display organ values, thresholds, and abnormalities.

10	Graphs display minimum and maximum thresholds for each organ to highlight abnormalities.	Viewing Results of Scan	ScanPlotDialog, QCustomPlot	Verify that threshold lines are displayed correctly on the graph.
11	Scan results include a body visualization with color-coded organ performance levels.	Viewing Results of Scan	ScanPlotDialog	Verify the body visualization and color codes (green, blue, red).
12	Users can view recommendations based on scan results.	Viewing Results of Scan	ScanPlotDialog	Verify that actionable recommendations are displayed for abnormal results.
13	Recommendations for scans are saved in the JSON file and persist across sessions.	Viewing Results of Scan	ScanPlotDialog, User	Add recommendations and verify they are stored in the JSON file.
14	Users can delete profiles along with their associated scan data.	Initial Setup/Onboarding	MainWindow, User	Delete a profile and verify the JSON file is updated accordingly.
15	The system alerts the user when the device battery is low.	Start Metering/Scanning	MainWindow	Verify low battery warnings appear during scans with low device charge.
16	If the device cannot pair, the app provides instructions for retrying.	Preparing/Setting Up for First Scan	MainWindow	Simulate pairing failure and observe the retry instructions.
17	Scan results can be filtered by date or user profile in the history section.	Viewing Results of Scan	MainWindow	Select a profile and scan date to verify filtering works as expected.
18	The app shuts down gracefully when the battery level is critically low.	Start Metering/Scanning	MainWindow	Simulate low battery and observe the shutdown process.